

CIFAR and Office-Home Diagnostic Experiments

Results for Plan Priorities 3–7

24 September 2026

Scope and main method. This report addresses exactly the experimentally assigned modules of the supplied plan: Office-Home Priorities 3–6 and CIFAR Priority 7. GMM, new benchmarks, paper changes, and the contingent intervention are outside this report. The Meta-Diff main method is **correct source + linear transport**, namely the correct Product/clean-source task coordinate transformed by the frozen learned affine map $\tilde{z}_T = Az_S + b$. All transport here is linear/affine; direct z_T generation and the explicit class-embedding model are *diagnostics*, not alternate claims of zero-shot superiority. The frozen Office-Home and CIFAR EMA checkpoints were unchanged. Results are exploratory and must not replace the historical headline benchmark.

How to read contrasts and stars. Before every contrast table below, a positive-value rule states which route it favors. A star marks a 95% bootstrap interval that excludes zero, using unrounded limits; it is not a multiple-comparison-adjusted p -value. Office-Home and generation/time-bin intervals resample classes (paired where the same class is compared). CIFAR retrieval intervals resample 20 fixed support sets *conditional on the same fixed classes*; their stars do not establish population-level class significance. Lower DINO sliced-Wasserstein (SW), DINO-MMD, and denoising MSE are better; higher retrieval, requested-class identity, and Clipart share are better. Clipart share measures domain, not object identity. Real-image requested-class accuracy calibrates the semantic evaluator.

Priority 3: Office-Home transported-coordinate retrieval

Question. Does the transported Product coordinate retrieve its own held-out Clipart task, and is this better than raw source, wrong-source transport, and relation-only coordinates?

Ten held-out classes, 20 fixed support resamples, 16 Product source images and at most 32 disjoint Clipart target-support images per class; retrieval is among ten target-class coordinates. A positive margin means the correct target is nearer than the nearest wrong target (L2), or has greater similarity than the nearest wrong target (cosine).

Query route	L2 top-1 (%)	Cosine top-1 (%)	L2 margin	Cosine margin
Raw z_S	12.00	26.50	−0.05002	−0.17493
correct source + linear transport	42.00	44.50	−0.01528	−0.06195
Wrong-source transport	20.00	19.00	−0.08968	−0.26536
Relation only	10.00	10.00	−0.05448	−0.15104

Contrast sign: main minus comparator; positive favors correct source + linear transport. Entries are percentage-point top-1 differences with paired class-cluster bootstrap 95% intervals.

Comparator	L2 Δ [95% CI]	Cosine Δ [95% CI]
Raw z_S	+30.00 [+4.00, +58.50]*	+18.00 [−2.50, +43.50]
Wrong-source transport	+22.00 [−11.50, +56.00]	+25.50 [−10.00, +60.00]
Relation only	+32.00 [+5.50, +59.50]*	+34.50 [+7.00, +61.50]*

Answer. Transport improves target-coordinate retrieval over untransported source in L2, but the correct-versus-wrong-source class interval overlaps zero; therefore the data do not establish a statistically secure source-specific advantage on these ten Office-Home classes. The main route is still below 50% top-1 and has a negative mean margin. Relation-only queries are identical across classes, so their 10% is a tie/chance control, not identity retrieval. Coordinate alignment is imperfect before generation.

Priority 4: Office-Home seen-task versus unseen-task direct- z_T generation

Question. If transport is bypassed and an abundant target coordinate is supplied, does the frozen generator produce the requested object on meta-training classes and/or held-out classes?

For each class, all available Clipart support images outside 20 fixed real queries were encoded as z_T . The frozen EMA model generated 96 images/class using 50-step DDIM. “Seen” means the class was used in original Meta-Diff training; the seen-class query images were not necessarily excluded from that original training. This module deliberately does *not* evaluate the main zero-shot route, because its purpose is to isolate generative use of a target coordinate.

Class group	Classes	Generated identity (%)	Real identity (%)	DINO-SW ↓	DINO-MMD ↓	Clipart (%)
Seen	45	2.38	50.33	0.03679	0.35175	87.34
Held-out	10	1.56	52.00	0.03902	0.40642	88.33

Contrast sign: held-out minus seen; positive favors held-out classes, not the main method. The groups contain different classes, so this is an independent-class descriptive contrast, not a paired causal effect.

Contrast	Generated identity Δ [95% CI]
Held-out minus seen	−0.82 pp [−2.45, +0.76]

Answer. Requested-object identity is near absent in *both* class groups despite abundant true Clipart support and high Clipart-domain share. There is no demonstrated seen-to-unseen identity drop by this class-bootstrap interval. Since transport is bypassed, transport alone cannot explain the semantic failure. The result implicates the learned $z \rightarrow$ generation path or the present generative/evaluation regime, but does not uniquely identify the U-Net architecture as the cause.

Priority 5: Office-Home coordinate usefulness across diffusion time

Question. Does the frozen denoiser use the correct direct target coordinate, especially at high noise, and does use change between seen and held-out classes?

On real Clipart queries, each correct z_T was paired with a cyclic wrong-class z_T at identical (x_t, ϵ, t) . Each row below gives the *absolute* correct and wrong ϵ -MSE and their paired difference. Bins contain 200 diffusion steps; 800–999 is the highest-noise bin. Intervals resample classes.

Contrast sign: $\Delta_z = \mathcal{L}_\epsilon(z_{\text{wrong}}) - \mathcal{L}_\epsilon(z_{\text{correct}})$; positive favors the correct target coordinate, not specifically transport.

Group	Timesteps	Correct MSE	Wrong MSE	Δ_z [95% CI]
Seen	0–199	0.056513	0.056724	+0.000211 [+0.000141, +0.000294]*
	200–399	0.019655	0.019785	+0.000131 [+0.000086, +0.000183]*
	400–599	0.013028	0.013164	+0.000136 [+0.000089, +0.000189]*
	600–799	0.008825	0.008963	+0.000138 [+0.000095, +0.000187]*
	800–999	0.003287	0.003342	+0.000055 [+0.000039, +0.000072]*
Held-out	0–199	0.059910	0.059928	+0.000018 [−0.000044, +0.000073]
	200–399	0.022160	0.022180	+0.000020 [−0.000070, +0.000098]
	400–599	0.014911	0.014948	+0.000037 [+0.000006, +0.000068]*
	600–799	0.009441	0.009505	+0.000063 [+0.000006, +0.000144]*
	800–999	0.003140	0.003157	+0.000017 [+2.15 × 10 ^{−7} , +0.000034]*

Answer. The correct coordinate slightly lowers denoising MSE, so the generator is not strictly coordinate-blind. Its held-out advantage is much weaker than on seen classes and is very small at highest noise, where generation must recover structure from nearly pure noise. This is consistent with, but does not prove, inadequate semantic use of z during generation. Denoising-loss differences alone do not quantify requested-object fidelity.

Priority 6: Explicit class-conditioning capacity control

Question. Can the matched U-Net/FiLM generative path express semantic object identity when given explicit class labels rather than inferred z or transport?

A newly initialized, otherwise matched 64-pixel FiLM U-Net received learned 16-dimensional embeddings for 45 training Clipart classes. It used neither set encoder nor transport; it completed 50,000 updates with 64 independently noised observations/update. Evaluation held out the same 20 query-image IDs per class used for Priority 4, with 96 generated images/class and 50-step DDIM. This is a capacity diagnostic, not a Meta-Diff baseline or a zero-shot comparison.

Seen-class route	Generated identity (%)	Real identity (%)	DINO-SW ↓	DINO-MMD ↓	Clipart (%)
Original direct z_T	2.38	50.33	0.03679	0.35175	87.34
Explicit class embedding	6.90	50.33	0.03507	0.32768	87.45

Contrast sign: identity and Clipart share are explicit embedding minus direct z_T ; SW and MMD are direct z_T minus explicit embedding. Positive therefore favors the explicit class control, not the main method. Intervals are paired by the 45 classes.

Metric	Favor-control Δ [95% CI]
Requested-class identity	+4.51 pp [+2.85, +6.37]*
DINO-SW reduction	+0.00172 [+0.00121, +0.00229]*
DINO-MMD reduction	+0.02407 [+0.01332, +0.03671]*
Clipart-share increase	+0.12 pp [−1.30, +1.53]

Answer. Explicit conditioning produces a measurable semantic improvement, so the FiLM interface is not wholly incapable of using labels. However, 6.90% identity remains far below the 50.33% real-query evaluator calibration and does not demonstrate reliable class generation. Thus this control does not establish that the backbone under matched training can generate semantic classes successfully; the plan’s prerequisite for a high-noise ranking intervention is not met, and no such run was performed. The original model may have trained on seen-class query images, whereas this new control explicitly withheld them; this difference limits a causal attribution of the gain.

Priority 7: Matching CIFAR diagnostics

Question. Do the same coordinate-retrieval, direct- z_T generation, and timestep probes explain why controlled CIFAR distributional transport is stronger than semantic class generation?

The frozen 50k EMA FiLM + affine-transport model was evaluated on all three original corruption relations (Gaussian blur, Gaussian noise, contrast), 60 seen and 20 held-out fine classes. Retrieval uses 20 fixed 64-source/64-target support resamples, with candidates confined to the corresponding seen or held-out group. Direct- z_T generation uses 200 target support images and 50 disjoint real target queries/class after a post-hoc repartition, 96 generated samples/class and 50-step DDIM. A frozen DINO nearest-prototype instrument, calibrated on the real queries, ranks all 80 classes; its semantic percentages are diagnostic, not historical headline metrics.

7a. Transported-coordinate retrieval

Each table cell below is *top-1 percentage / mean nearest-wrong margin*; a positive margin favors the correct class. “Main” is correct source + linear transport. Relation-only coordinates are class-constant, so their chance/tie behavior is not semantic retrieval.

Group	Relation	Metric	Raw z_S	Main	Wrong-source + transport	Relation only
Seen	Blur	L2	36.50/ − 0.01050	71.50/ + 0.02241	0.42/ − 0.24324	1.67/ − 0.13592
		Cosine	26.58/ − 0.06849	72.58/ + 0.01675	0.42/ − 0.24228	1.67/ − 0.10330
Seen	Noise	L2	56.83/ + 0.00548	74.50/ + 0.02872	0.33/ − 0.26493	1.67/ − 0.13956
		Cosine	30.50/ − 0.04843	78.08/ + 0.01791	0.25/ − 0.24565	1.67/ − 0.09906
Seen	Contrast	L2	36.00/ − 0.01447	68.83/ + 0.02719	0.50/ − 0.26657	1.67/ − 0.12781
		Cosine	25.58/ − 0.07178	72.08/ + 0.02048	0.67/ − 0.28837	1.67/ − 0.11249
Held-out	Blur	L2	17.25/ − 0.02427	62.50/ + 0.01114	1.25/ − 0.12900	5.00/ − 0.08812
		Cosine	16.75/ − 0.13231	64.50/ + 0.00806	1.25/ − 0.11517	5.00/ − 0.07663
Held-out	Noise	L2	26.50/ − 0.01098	59.50/ + 0.00985	0.75/ − 0.11476	5.00/ − 0.08370
		Cosine	15.50/ − 0.09900	60.00/ + 0.00693	0.00/ − 0.09392	5.00/ − 0.06372
Held-out	Contrast	L2	14.00/ − 0.04431	49.00/ − 0.00423	2.25/ − 0.09975	5.00/ − 0.07584
		Cosine	10.25/ − 0.16097	48.25/ − 0.00313	2.00/ − 0.09921	5.00/ − 0.07595

Contrast sign: main method minus comparator L2 top-1; positive favors correct source + linear transport. Intervals pair the same 20 support resamples of the *fixed* class set; stars indicate stability to support choice only, not independent-class significance.

Group	Relation	Main − raw source (pp) [95% CI]	Main − wrong source (pp) [95% CI]
Seen	Blur	+35.00 [+32.25, +37.67]*	+71.08 [+69.08, +73.08]*
Seen	Noise	+17.67 [+15.33, +20.00]*	+74.17 [+72.25, +75.92]*
Seen	Contrast	+32.83 [+30.67, +35.08]*	+68.33 [+66.58, +70.08]*
Held-out	Blur	+45.25 [+42.25, +48.25]*	+61.25 [+58.25, +64.00]*
Held-out	Noise	+33.00 [+27.25, +38.25]*	+58.75 [+54.50, +63.25]*
Held-out	Contrast	+35.00 [+30.25, +39.50]*	+46.75 [+43.00, +50.50]*

Retrieval answer. On all three held-out relations, main-method target-coordinate retrieval is much stronger than raw source and wrong-source transport across fixed support resamples. Blur/noise held-out margins are positive, but contrast’s held-out mean margins remain slightly negative. This is evidence of a useful, source-specific transported coordinate, not evidence that generated images preserve object class.

7b. Seen versus unseen direct- z_T generation

As in Priority 4, this deliberately bypasses the main method’s transport to test the generator’s use of a well-estimated target coordinate. The raw values and real-image instrument calibration are:

Group	Relation	Generated identity (%)	Real identity (%)	DINO-SW ↓	DINO-MMD ↓
Seen	Blur	5.52	50.80	0.02176	0.15937
Seen	Noise	3.78	29.27	0.01671	0.09777
Seen	Contrast	6.84	78.10	0.02983	0.27962
Held-out	Blur	3.59	51.20	0.02338	0.18884
Held-out	Noise	3.85	34.50	0.01888	0.13402
Held-out	Contrast	2.71	70.90	0.03059	0.28967

Contrast sign: held-out minus seen generated identity; positive favors held-out classes, not the main method. Intervals independently resample the different seen and held-out class sets.

Relation	Identity Δ (pp) [95% CI]
Blur	−1.93 [−6.70, +4.11]
Noise	+0.07 [−4.34, +6.18]
Contrast	−4.13 [−7.99, +0.14]

Generation answer. Direct abundant z_T gives very low class identity even on seen classes, far below the relation-specific real-image evaluator ceilings. None of the seen/unseen identity contrasts excludes zero. Thus semantic generation can fail after a correct target coordinate has already been provided; a transport-only explanation is insufficient. These post-hoc direct- z_T numbers are not directly comparable with historical zero-shot generation scores because support/query partition and identity instrument differ.

7c. Correct-versus-wrong target coordinate across diffusion time

Absolute ϵ -MSEs use identical real target query, noise, and timestep in each pair. Intervals resample classes. The 800–999 bin is highest noise. All values below are unscaled MSE.

Contrast sign: wrong-coordinate MSE minus correct-coordinate MSE; positive favors the correct target coordinate, not specifically transport. A star indicates an unrounded class-bootstrap 95% interval excluding zero. Cells with limits rounded to 0.00000 were starred from the unrounded limits.

Group	Relation	Timesteps	Correct MSE	Wrong MSE	Δ_z [95% CI]
Seen	Blur	0–199	0.069118	0.069843	+0.00073 [+0.00058, +0.00089]*
		200–399	0.025385	0.026199	+0.00081 [+0.00065, +0.00099]*
		400–599	0.014950	0.015695	+0.00074 [+0.00061, +0.00089]*
		600–799	0.008320	0.008864	+0.00054 [+0.00045, +0.00065]*
		800–999	0.002333	0.002566	+0.00023 [+0.00019, +0.00028]*
Seen	Noise	0–199	0.575592	0.576098	+0.00051 [+0.00040, +0.00062]*
		200–399	0.149464	0.150564	+0.00110 [+0.00087, +0.00135]*
		400–599	0.055730	0.056779	+0.00105 [+0.00083, +0.00129]*
		600–799	0.020958	0.021725	+0.00077 [+0.00063, +0.00092]*
		800–999	0.004284	0.004563	+0.00028 [+0.00023, +0.00033]*
Seen	Contrast	0–199	0.097140	0.098230	+0.00109 [+0.00084, +0.00137]*
		200–399	0.025000	0.026042	+0.00104 [+0.00082, +0.00130]*
		400–599	0.011904	0.012583	+0.00068 [+0.00055, +0.00083]*
		600–799	0.005251	0.005642	+0.00039 [+0.00032, +0.00047]*
		800–999	0.001187	0.001322	+0.00014 [+0.00011, +0.00016]*
Held-out	Blur	0–199	0.079225	0.079248	+0.00002 [−0.00008, +0.00012]
		200–399	0.030886	0.030951	+0.00006 [+0.00000, +0.00013]*
		400–599	0.018033	0.018145	+0.00011 [+0.00005, +0.00018]*
		600–799	0.009747	0.009893	+0.00015 [+0.00005, +0.00026]*
		800–999	0.002668	0.002732	+0.00006 [+0.00002, +0.00012]*
Held-out	Noise	0–199	0.584385	0.584397	+0.00001 [−0.00004, +0.00006]
		200–399	0.157766	0.157801	+0.00004 [−0.00005, +0.00013]*
		400–599	0.060890	0.060990	+0.00010 [+0.00005, +0.00016]*
		600–799	0.023057	0.023172	+0.00012 [+0.00004, +0.00021]*
		800–999	0.004634	0.004695	+0.00006 [+0.00002, +0.00011]*
Held-out	Contrast	0–199	0.111840	0.111861	+0.00002 [−0.00011, +0.00016]
		200–399	0.030179	0.030231	+0.00005 [−0.00006, +0.00015]
		400–599	0.013888	0.013959	+0.00007 [−0.00000, +0.00015]
		600–799	0.005905	0.005997	+0.00009 [+0.00003, +0.00017]*
		800–999	0.001368	0.001395	+0.00003 [+0.00001, +0.00005]*

Time-bin answer. Correct-coordinate denoising advantages are positive across all seen cells, but smaller and sometimes uncertain on held-out classes. They are not zero at highest noise, yet their absolute magnitude there is small. Alongside strong coordinate retrieval and poor direct- z_T object identity, this suggests a bottleneck in *semantic use* of the coordinate after transport, not simply loss of target identity in the affine map. The probes do not uniquely blame the U-Net architecture, and they do not constitute a rerun of the CIFAR headline suite.

Audit and interpretation boundary. The completed-run audit verified 20 retrieval resamples, all 45/10 Office and 60/20 CIFAR seen/held-out classes, three CIFAR relations, five time bins, 96 generated images/class, 50-step DDIM, 50,000 capacity updates with effective 64 independent noise observations, disjoint diagnostic target support/queries, and frozen checkpoint/manifest hashes. Office checkpoint SHA-256 begins 44e2710a; CIFAR checkpoint SHA-256 begins f7c1369c. The audit and report source are in `artifacts/cifar_office_priority_diagnostics_v1/`; CIFAR raw data are in `artifacts/cifar_priority7_v1/`. Office raw and capacity outputs are in the experiment’s `priority_diagnostics_v1` and `class_capacity_v1` directories. All identity claims are bounded by post-hoc evaluator calibration; stars are exploratory and unadjusted. Correct-source transport is advantageous for coordinate retrieval, especially on CIFAR, but these experiments do *not* demonstrate semantic-generation superiority of the main method over every alternative.